

We noted that most of the high-energy error MOF structures have high atomic density (Figure S41c inset), and that the errors cannot be canceled by adding dispersion correction. In Figure S41b, we analyze the element-wise energy error per atom by distributing the absolute energy error per atom to the constituent elements by the corresponding composition in each MOF structure. As presented in Figure S41b, there is a strong elemental dependence of energy error per atom. Most of high error elements can be attributed to the difference of chosen pseudopotentials used by MP and QMOF databases (in particular the choices of which electrons are treated as valence states), see Table S3.

Figure S41c and Figure S41d present the incompatible QMOF entries excluded from the analysis in Figure 3a. The incompatibility stems from (i) different, incompatible pseudopotentials (PSPs), or (ii) mismatched calculation parameters (with or without Hubbard U correction) used in QMOF and MP database, or the combinations of both. Figure S41c compares MACE energies with QMOF PBE energies calculated with incompatible PSPs (case i). Figure S41d compares MACE energies with QMOF PBE calculations on structures that would be calculated at PBE+U level by MP standard at the curation time of MPTrj.

MPtrj	Be	Bi	Cu_pv	Eu	Fe_pv	Gd	Li_sv	Mg_pv	Mo_pv	Nb_pv	Ni_pv	Os_pv	Re_pv	Ti_pv	V_pv	W_pv	Yb_2
QMOF	Be_v	Bi_d	Cu	Eu_3	Fe	Gd_3	Li	Mg	Mo_sv	Nb_sv	Ni	Os	Re	Ti_sv	V_sv	W_sv	Yb_3

Table S3: Difference in VASP POTCARs used by MPTrj and QMOF

A.28.2 CO₂ adsorption

All the calculations for the CO₂ dynamics with MOFs were performed with MACE-MP-0 by adding the D3 dispersion correction (29) to the MACE-MP-0 potential. The simulations were carried out with ASE (275) on a cell containing 165 atoms with one CO₂ molecule, initialised at the centre of the pore, using NVT Langevin dynamics (276) with a friction factor of $5 \times 10^{-3} \text{ fs}^{-1}$. The temperature was set to 600 K with a time step of 1 fs. Twenty-four 1 ns trajectories were generated using different initial velocities, and all quantities presented were averaged over all of them, discarding the first 2 ps from each to account for equilibration. All structures had their cells and positions optimised at start using `FrechetCellFilter` from ASE. The code used to generate the trajectories is available in the repo (208) and (277).

Similarity statement

The MP dataset does not contain MOFs. There are 8 structures containing all MgOCH elements, and 62 structures that have MgOHC elements on their own or along with other elements. Based on UMAP analysis, we see that most atomic environments, both MgO and linkers, in the example system, are similar to environments in the training data but none is Mg-MOF-74 specific.

The closest (most relevant) structures in the training set are CO₂ (mp-556034 mp-20066 mp-995224 mp-11725 mp-644607 mp-1102227 mp-1190685 mp-995198 mp-1190699 mp-1077906 mp-1077316 mp-729728). CO₂ alone matches 4896 structures with C, O and alongside other elements.

We provide

- `co2_FilterType.exclusive_OC_chemiscope_input.json`
- `mg-mof-74-co2_FilterType.exclusive_MgOCH_chemiscope_input.json`
- `mg-mof-74-co2_FilterType.exclusive_MgOCH_chemiscope_input.json`

that contain exact matches of Mg, O, C and H and the inclusive versions

- `mg-mof-74-co2_FilterType.exclusive_MgOCH_chemiscope_input.json`
- `mg-mof-74-co2_FilterType.inclusive_MgOCH_chemiscope_input.json`
- `mg-mof-74_FilterType.inclusive_MgOCH_chemiscope_input.json`

to help visualize the interactive UMAP on chemiscope.org.

Performance summary

Excellent energy prediction for large database of MOFs. Correct prediction of binding structure and free energy of CO₂ in Mg-MOF-74.

A.29 Combinatorial Materials Discovery

A.29.1 Formation energy of hypothetical materials

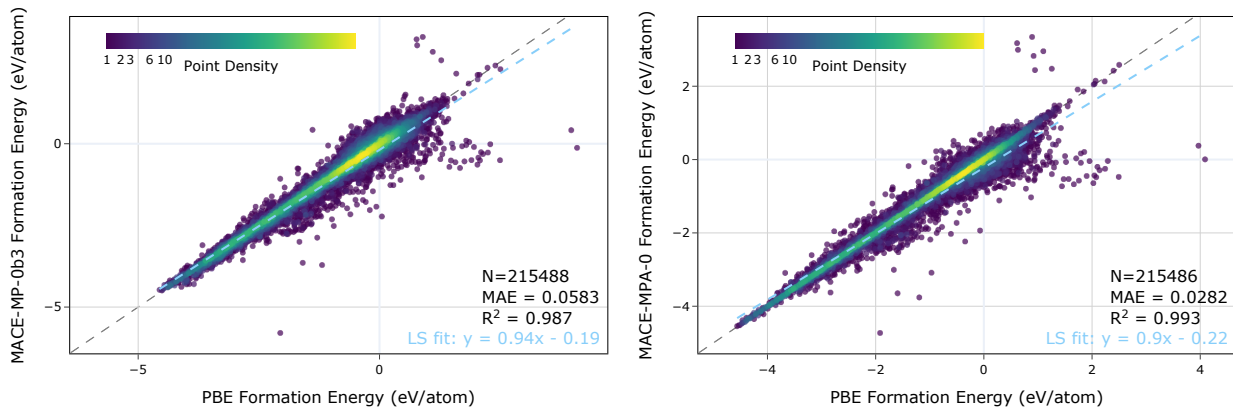


Figure S42: Formation energy parity plot showing the difference between the DFT-relaxed energy and the MACE-MP-0b3-relaxed energy (left) and MACE-MPA-0-relaxed energy starting (right) from WBM initial structures.

MACE-MP-0b3 trained on the MPtrj dataset and MACE-MPA-0 trained on the combined MPtrj and sAlex (111) datasets generalize to out-of-distribution (OOD) chemistries as shown by their performance on the WBM dataset (278) which was generated using elemental substitutions drawn according to a data-mined chemical similarity measure (279). The initial set of 9,524 structures for substitution were taken from the Materials Project convex hull. The substituted structures were relaxed using the MPRelaxSet PBE DFT workflow. After this, the convex hull was recalculated using the new structures and then subsequent rounds of substitution were carried out on the new structures that ended up on the combined MP plus growing WBM convex hull. In total, 5 rounds of substitutions were carried out yielding a dataset of 257,487 inorganic crystals that are OOD with respect to MP and therefore well suited to benchmarking. For this investigation, we use the cleaned version of the WBM dataset released in Matbench Discovery (MBD) (280), which first discards 524 crystals with unphysical or missing labels. Subsequently, all structures in WBM with composition+prototype matching a structure in MP are removed. Within WBM composition+prototype, duplicates are dropped leaving only the lowest energy structure. The final test set consists of 215 488 materials, and of these we obtained MACE-MP-0b3 relaxed structures for 215 488 and 215 486 for MACE-MPA-0.

Following the MBD protocol, we use MACE-MP-0b3 to relax the initial substituted structures and compare these predictions against the ground truth formation energy calculated with DFT. The predictions on the MACE-MP-0b3 self-relaxed structures result in an MAE of 58 meV/atom as shown in Fig. S42. MACE-MPA-0 cuts this error more than in half, achieving a formation energy MAE of 28 meV/atom. Similarly, MACE-MPA-0’s R^2 improves on MACE-MP-0b3’s R^2 , going from 0.987 to 0.993. However, both model’s energy error exhibit a systematic tendency to underpredict the PBE formation energy as seen in the least-square fits of $y = 0.94x - 0.19$ for MACE-MP-0b3 and $y = 0.9x - 0.22$ for MACE-MPA-0. Interestingly, despite the lower error and higher R^2 , MACE-MPA-0’s error is more systematic since slope and intercept are further from 1 and 0, respectively. Thus a simple affine fit to correct for the systematic component in the energy error would reduce MACE-MPA-0’s already lower error more than MACE-MP-0b3’s. While Alexandria is overall a highly valuable training set, this result is consistent with the observation that it is even more biased towards low-energy structures than MPtrj.

The OOD nature of the WBM dataset can be seen in the increase in MAE between WBM batches from 48 meV/atom for the first batch to 78 meV/atom for the final batch as the structures become increasingly dissimilar to MP due to accumulated substitutions. When these predictions are used to attempt to classify whether the structure lies above or below the MP convex hull training set, MACE-MP-0b3 achieves an F1 score of 0.66 and a discovery acceleration factor (DAF) of 3.95 compared to F1 of 0.85 and DAF of 5.58

for MACE-MPA-0. The DAF is the ratio of the precision (TP/PP) to the prevalence (P/N) of the test set. Here, TP = True Positives, PP = Predicted Positives, P = Total Positives and N is the test set size. These results show that MACE-MP-0b3 and especially MACE-MPA-0 can extrapolate to novel chemistries and are well-suited to high-throughput materials discovery.

Similarity statement

By construction there is no overlap in terms of both composition and prototype together between MP and the WBM test set studied here. However there is overlap for both compositions and prototypes separately. MP contains 105 583 unique reduced formulae, whilst the WBM test set contains 160 055. Of these 15 782 overlap MP albeit with all instances being examples of different prototype structures. MP contains 32 933 different isopointal prototypes, whilst the WBM test set contains just 2816. Of the prototypes found in WBM 1813 are also found in MP. Of the isopointal prototypes not seen in MP the 5 most common are (occurrences in parentheses): ABC2_oI8_71_a_b_f (323), ABC2_hP12_181_c_d_i (215), AB2_hP9_189_f_adg (156), ABC2_oI8_44_a_b_c (117), AB4C6_mC22_8_a_2ab_2a2b (106). These arise due to changes in symmetry during the relaxation of the substituted structures.

A.29.2 Stoichiometric substitutions

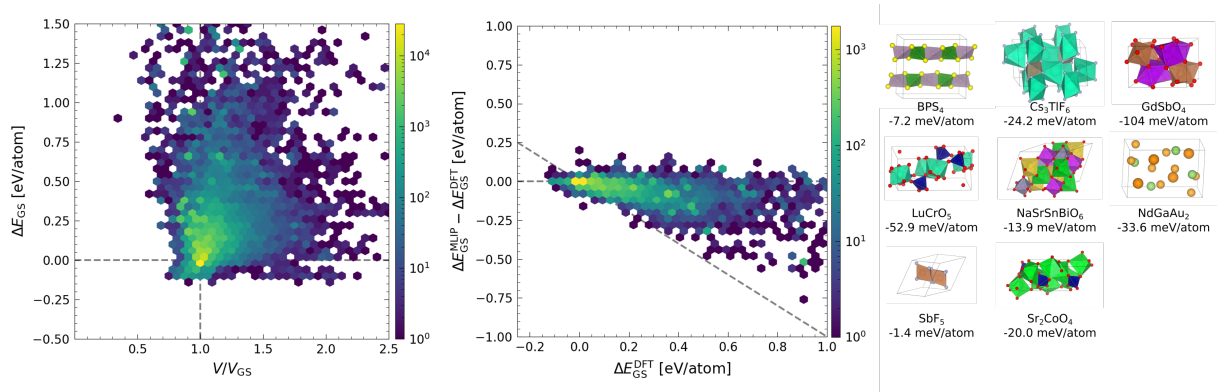


Figure S43: Left: Relative energies vs. volumes of ca. 150k relaxed structures generated via exhaustive element substitutions for 100 MP compositions. Center: Error of relative energies calculated with PBE and MACE-MP-0b3 for 6909 randomly drawn structures from the left plot, as a function of the PBE relative energy. Right: DFT validated examples of newly discovered stable phases, along with their energies relative to the MP convex hull.

To test the interpolative and extrapolative capabilities of the model within MP chemistry, exhaustive element substitutions for 100 randomly selected compositions were performed. Specifically, MP was first filtered to remove all compositions with more than 16 atoms in the reduced formula, to ensure that a sufficient number of possible substitutions. This set was randomly split into host and target compositions with a 60:40 ratio. 100 compositions were randomly drawn from the target split and substituted into each stoichiometrically matching host, yielding 154,685 substituted structures. These were optimized with MACE-MP-0 using full unit cell relaxations (without the D3 correction). In Fig. S43, the results are shown in terms of energies and volumes relative to the most stable structure of that composition within the MP. The distribution is sharply peaked at $\frac{V}{V_{ref}} = 1$ and $E - E_{ref} = 0$, indicating that the substituted cells often relax back to the known ground state structure from MP. This makes MACE-MP-0 suitable for predicting the crystal structures of unknown materials. For a random sample of 16126 structures, the MACE-MP-0 relative energies were validated with MP-compatible PBE DFT calculations, yielding an RMSE of 0.097 eV/atom.

Beyond recovering the MP ground state, the wide range of substitutions tested also yields a large number of alternative structures for each composition. 3522 of these have relative energies of zero or lower and are thus predicted to be more stable than the MP reference. To validate these predictions, DFT relaxations were performed for all structures with negative relative energies. The results confirms that 2724 of them are indeed more stable than the corresponding MP reference. Importantly, not each of these is a new stable phase, as some relaxations converged to the same minimum and multiple lower energy structures are found for some compositions. Nonetheless, for eight of the 100 compositions considered here, new structures with energies below the current MP hull were discovered (for BPS_4 , Cs_3TlF_6 , $GdSbO_4$, $LuCrO_5$, $NaSrSnBiO_6$, $NdGaAu_2$, SbF_5 , and Sr_2CoO_4). These constitute genuine predictions of new, thermodynamically stable phases relative to the MP convex hull. These phases were also not reported in the recent GNoME effort (which focused on compositions not included in MP). However, the WBM dataset does report slightly more stable structures for Cs_3TlF_6 and SbF_5 . It also includes structures for $NdGaAu_2$, which are less stable than the one reported herein.

Similarity statement

The MP contains at least one (*e.g.* $NaNiIO_6$, $NaSrSnBiO_6$) and at most 423 (Sr_2CoO_4) structures that contain all elements in the discovered stable phases. These matches are similar to the discovered structures, as host and reference structures are present in the MP. Yet, the discovered phases are unique and by definition not part of the training set. For each discovery, the closest structures in the training set are: mp-1106139